DATA TYPES

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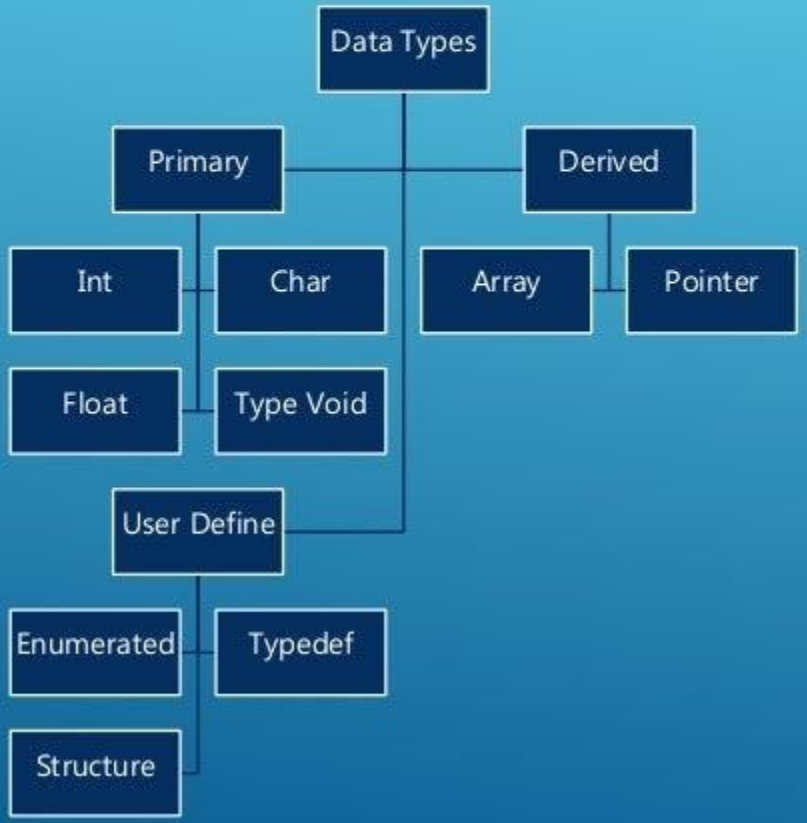
- In the C Programming language, data types refer to a broad system used for declaring variables or functions of different types. The type of a variable determines how much space it occupies in storage and how the bit pattern stored is interpreted.

TYPES OF DATA TYPES



INT DATA TYPE

- Integers are whole numbers with a range of values, range of values are machine dependent. Generally an integer occupies 2 bytes memory space and its value range limited to -32768 to +32767.

Data Type	Control String	Size	Range
int	%d	2 bytes	-32768 to +32767
long int	%ld	4 bytes	-2^{31} to $+2^{31}$
unsigned int	%u	2 bytes	0 to 65535

CHAR DATA TYPE

- Character type variable can hold a single character. As there are singed and unsigned int (either short or long), in the same way there are signed and unsigned chars; both occupy 1 byte each, but having different ranges. Unsigned characters have values between 0 and 255; signed characters have values from –128 to 127.

FLOAT DATA TYPE

- The float data type is used to store fractional numbers (real numbers) with 6 digits of precision. Floating point numbers are denoted by the keyword float. When the accuracy of the floating point number is insufficient, we can use the double to define the number. The double is same as float but with longer precision and takes double space (8 bytes) than float. To extend the precision further we can use long double which occupies 10 bytes of memory space.

Data Type	Size	Value Range	Precision
float	4 byte	1.2E-38 to 3.4E+38	6 Decimal Places
double	8 byte	2.3E-308 to 1.7E+308	15 Decimal Places
long double	10 byte	3.4E-4932 to 1.1E+4932	19 Decimal Places

TYPE VOID DATA TYPE

- The void type has no values therefore we cannot declare it as variable as we did in case of integer and float. The void data type is usually used with function to specify its type. Like in our first C program we declared "main ()" as void type because it does not return any value.

DERIVED DATA TYPE

- **Array:** An array in C language is a collection of similar data-type, means an array can hold value of a particular data type for which it has been declared. Arrays can be created from any of the C data-types int.
- **Pointer: C** Pointer is a special variable that can be used to store address of another variable.

ENUMERATED DATA TYPE (ENUM)

Enumerated data type is a user defined data type having finite set of enumeration constants. The keyword 'enum' is used to create enumerated data type. Enumeration data type consists of named integer constants as a list. It start with 0 (zero) by default and value is incremented by 1 for the sequential identifiers in the list.

Syntax:

Enum [data_type] {const1, const2… constn};

Enum example in C:

enum month { Jan, Feb, Mar }; or /* Jan, Feb and Mar variables will be assigned to 0, 1 and 2 respectively by default */

enum month { Jan = 1, Feb, Mar }; /* Feb and Mar variables will be assigned to 2 and 3 respectively by default */

enum month { Jan = 20, Feb, Mar }; /* Jan is assigned to 20. Feb and Mar variables will be assigned to 21 and 22 respectively by default */

TYPEDEF DATA TYPE

It is used to create new data type. But it is commonly used to change existing data type with another name.

Syntax:

typedef [data_type] new_data_type;

Example:

typedef int integer;

integer roll_no;

STRUCTURE DATA TYPE

C Structure is a collection of different data types which are grouped together and each element in a C structure is called member.

Example:

```
struct student {
    int roll_no;
    char name[20];
    char city[20];
}
```

THANK YOU